

Evaluating Matchmaking Fairness and Winning in Dota 2 by Bayesian: Skill or Chance?

Group 6: Jingyi Bai, Jiayi Tang, Ningyun Xie, Joy Wang



01

INTRODUCTION

State the research question and their significance





Why we choose this topic?



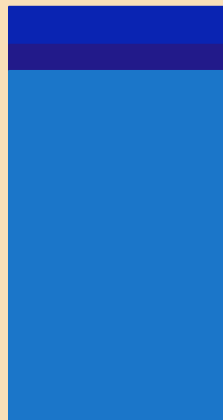
- **Topic:**
 - ◆ Evaluating Matchmaking Fairness and Winning factors in Dota 2.
- **Why:**
 - ◆ As esports grows, ensuring fairness in matchmaking is crucial for both players and developers.
- **Impact:**
 - ◆ Analyzing matchmaking fairness can enhance player experience by addressing potential flaws.
- **Data Availability:**
 - ◆ Dota 2 provides accessible, rich data, perfect for in-depth analysis of match factors.
- **Opportunity:**
 - ◆ Insights could lead to better matchmaking algorithms and player strategies.





02

DATA





DATA OVERVIEW



SOURCE

Kaggle Dota 2 Dataset



CRITERIA

Captains Game Mode
Excluding Anonymous
Players



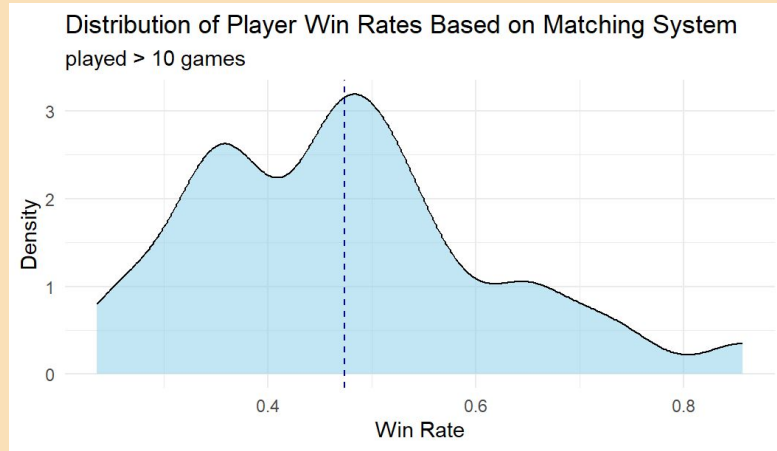
VARIABLES

Player-Level
Match-Level

	match_id	account_id	hero_id	duration	gold	teamfight_damage	teamfight_xp_end	xp_per_min	kills	deaths	assists
1	114	662	42	2714	3934	8489	95144	480	6	7	16
2	114	663	30	2714	2596	13178	75653	387	15	9	12
3	114	664	100	2714	3669	9928	124072	509	11	9	17
4	114	665	106	2714	6109	21277	184468	728	14	4	18
5	114	667	78	2714	1322	11190	118968	422	4	8	5

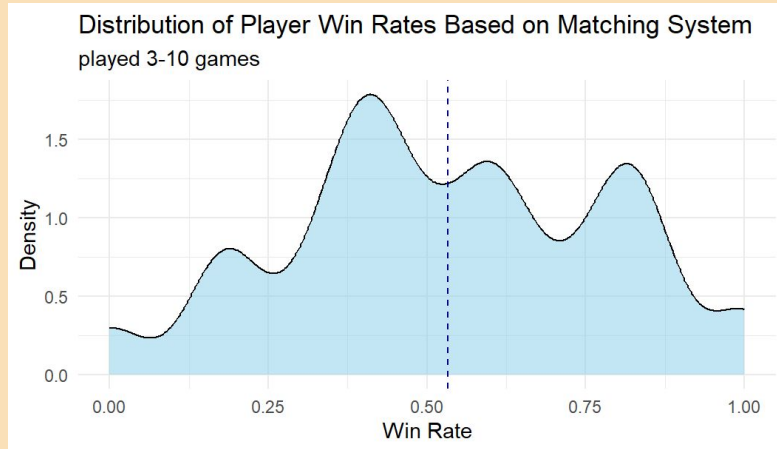


VISUALIZATION



→ **Win Rate Distribution (> 10 Games, n = 1246):**

- ◆ Win rates centered around 0.48
- ◆ indicating balanced matchmaking



→ **Win Rate Distribution (3-10 Games, n = 450):**

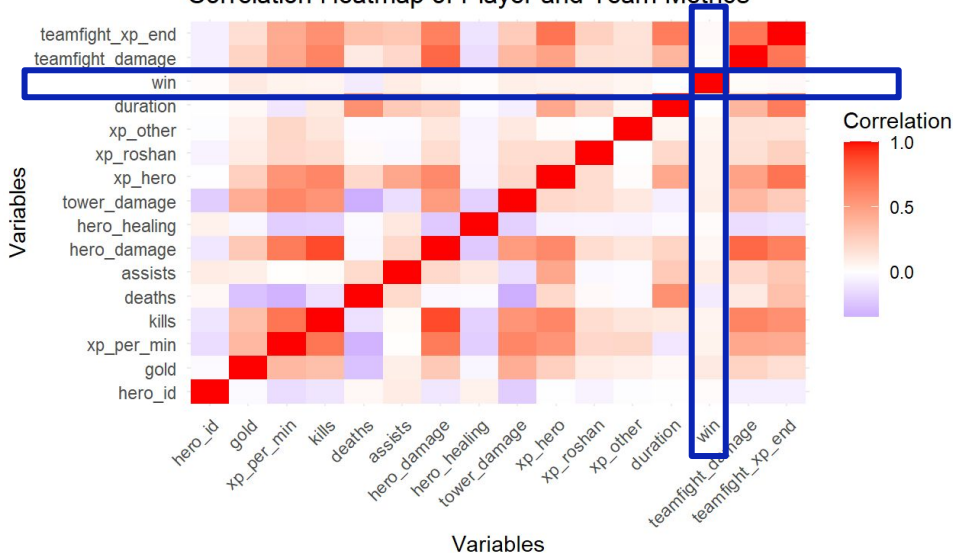
- ◆ Greater variance
- ◆ Possible imbalance for newer players.



VISUALIZATION



Correlation Heatmap of Player and Team Metrics



Correlation Heatmap



Correlations between kills, gold, and deaths

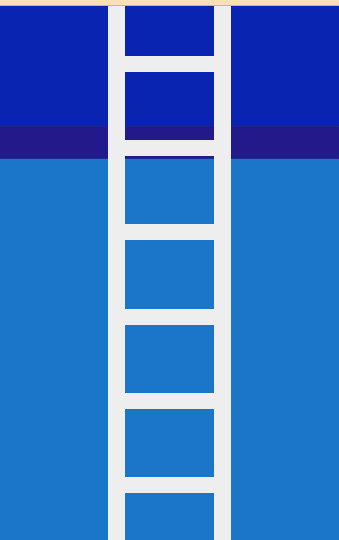
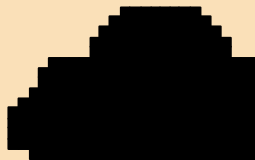


Highlighting their impact on match outcomes.



MODEL

Introduce notation, assumptions, and modeling choices



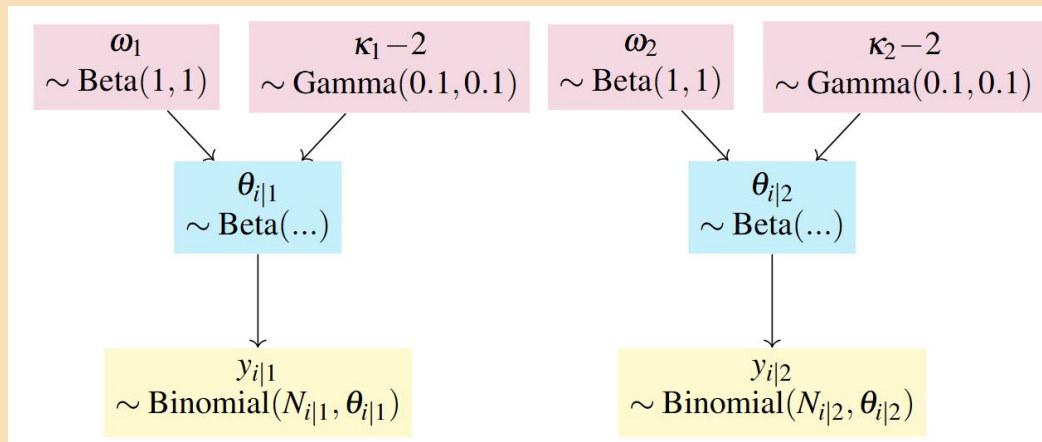


Hierarchical Beta-Binomial Model



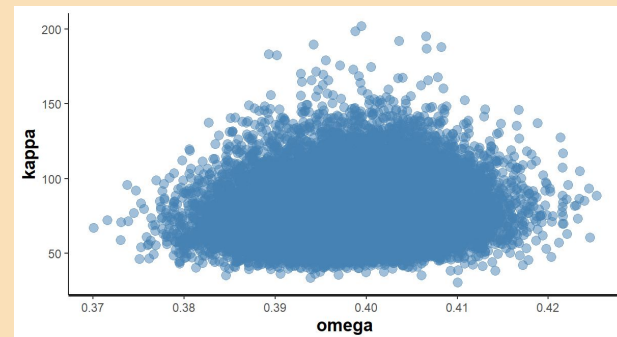
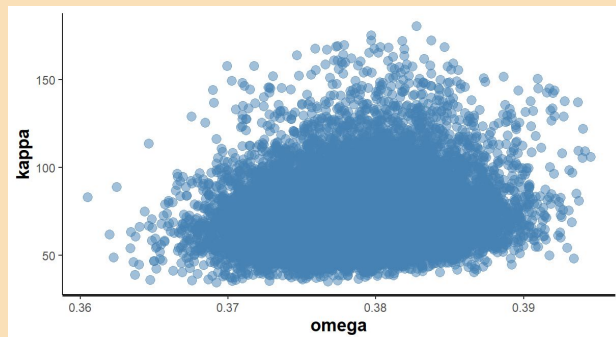
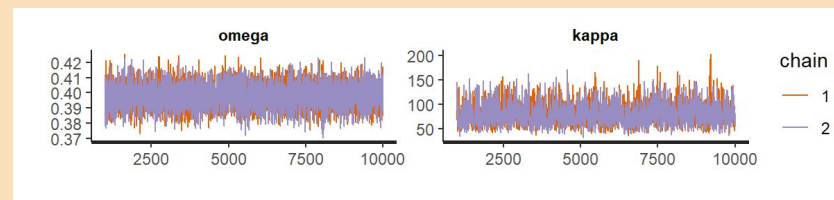
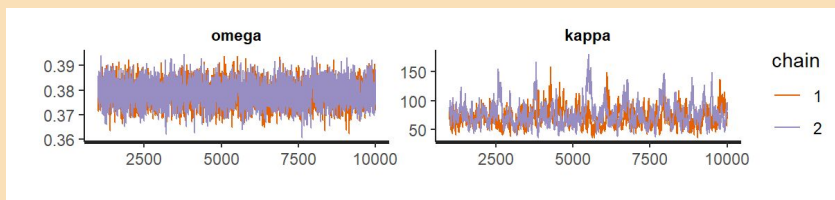
- Inexperienced players ($e = 1$): played between 3 to 10 games
- Experienced players ($e = 2$): played more than 10 games

$$\begin{aligned}y_{i|e} &\sim \text{Binomial}(N_{i|e}, \theta_{i|e}) \\ \theta_{i|e} &\sim \text{Beta}(\omega_e(\kappa_e - 2) + 1, (1 - \omega_e)(\kappa_e - 2) + 1) \\ \omega_e &\sim \text{Beta}(1, 1) \\ \kappa_e - 2 &\sim \text{Gamma}(1, 1)\end{aligned}$$



STAN: iteration = 10000, warmup = 1000, chain = 2

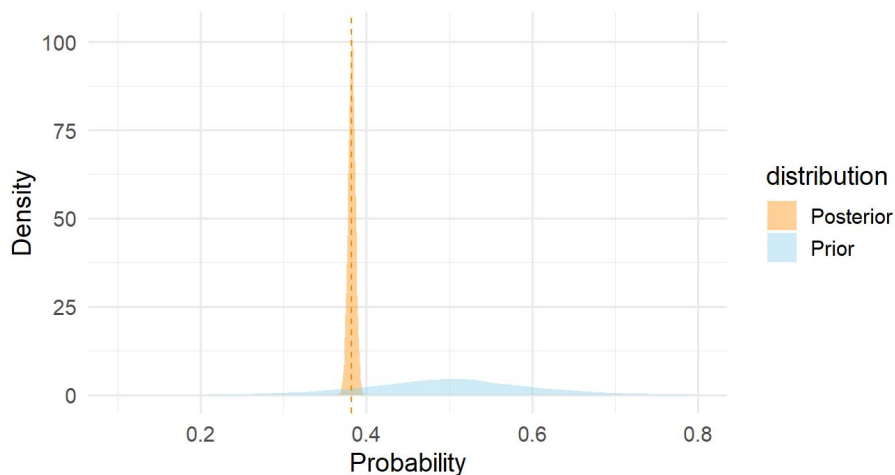
	Inexperienced players		experienced players	
	ω	$\kappa - 2$	ω	$\kappa - 2$
n_eff	2323	1721	3527	1135



Underlying "general winning rate" set by the match system for each category

$$E(\theta) = \frac{\alpha}{\alpha + \beta} = \frac{\omega(\kappa - 2) + 1}{\omega(\kappa - 2) + 1 + (1 - \omega)(\kappa - 2) + 1} = \frac{\omega(\kappa - 2) + 1}{\kappa}$$

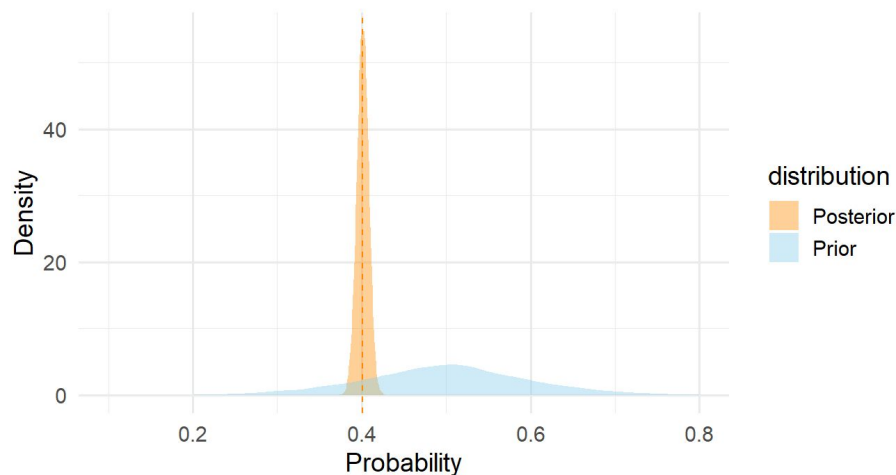
Prior and Posterior Distributions of $E(\theta_{i|1})$



Mean: 0.382

Credible Interval: (0.37, 0.39)

Prior and Posterior Distributions of $E(\theta_{i|2})$



Mean: 0.401

Credible Interval: (0.39, 0.41)



Logistic Regression



We apply logistic regression models to investigate the correlation between several performance factors and the outcome of a match.

We focus on individual player performance and team-level dynamics.



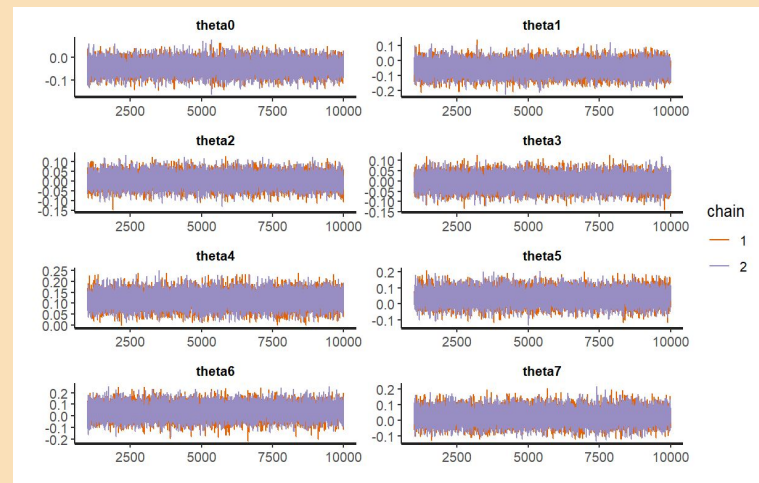
Logistic Regression Model - Individual

$$\log\left(\frac{P(y_i = 1)}{1 - P(y_i = 1)}\right) = \theta_0 + \theta_1 x_{1i} + \theta_2 x_{2i} + \theta_3 x_{3i} + \theta_4 x_{4i} + \theta_5 x_{5i} + \theta_6 x_{6i} + \theta_7 x_{7i}$$

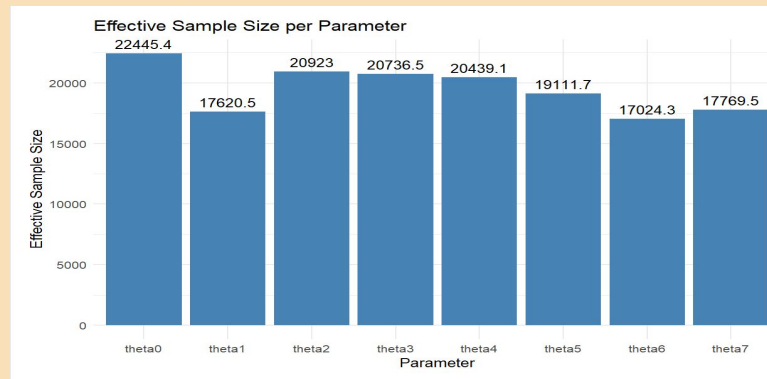
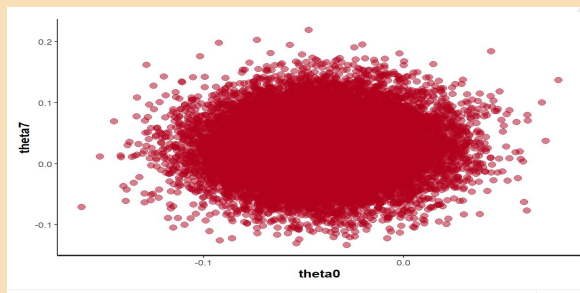
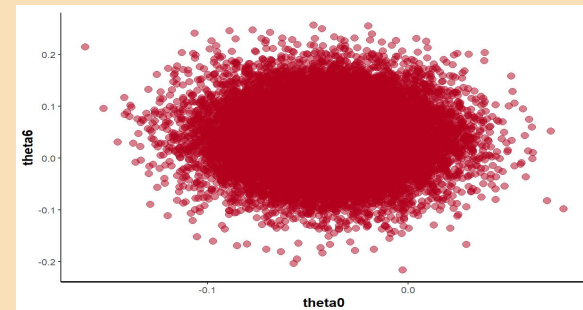
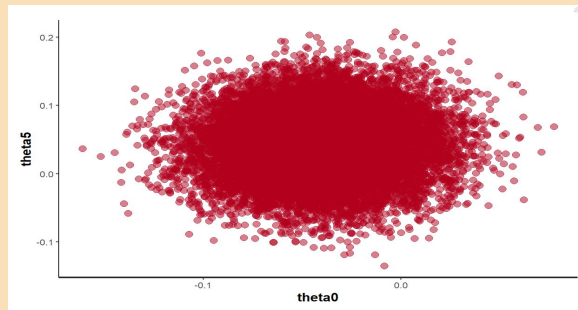
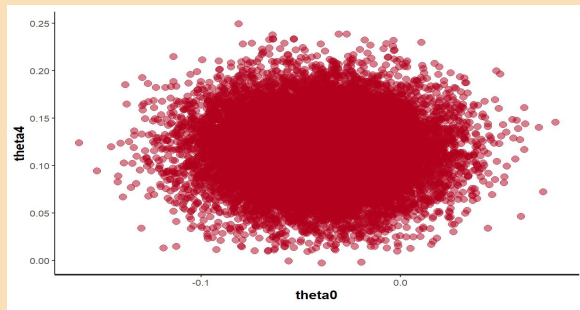
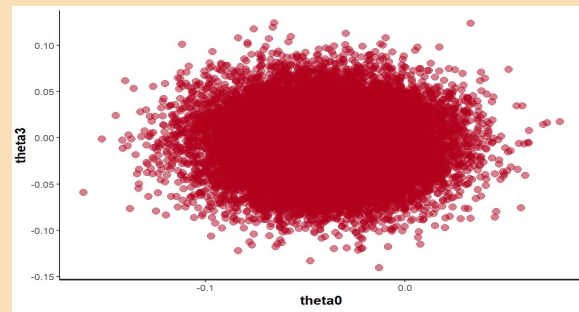
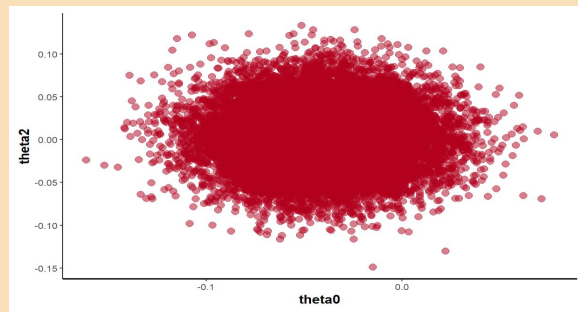
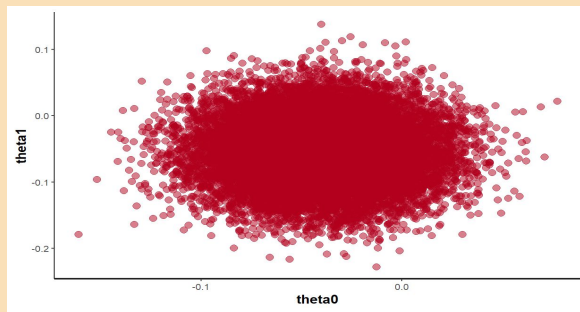
Variable and Coefficient Interpretations:

- θ_0 : Intercept term
- θ_1 : Effect of **kills**
- θ_2 : Effect of **deaths**
- θ_3 : Effect of **assists**
- θ_4 : Effect of **gold**
- θ_5 : Effect of **teamfight damage**
- θ_6 : Effect of **teamfight experience**
- θ_7 : Effect of **duration**

STAN: iteration = 10000, warmup = 1000, chains = 2



Each θ_j coefficient quantifies the log-odds change in the probability of winning associated with a one-unit increase in the corresponding variable.



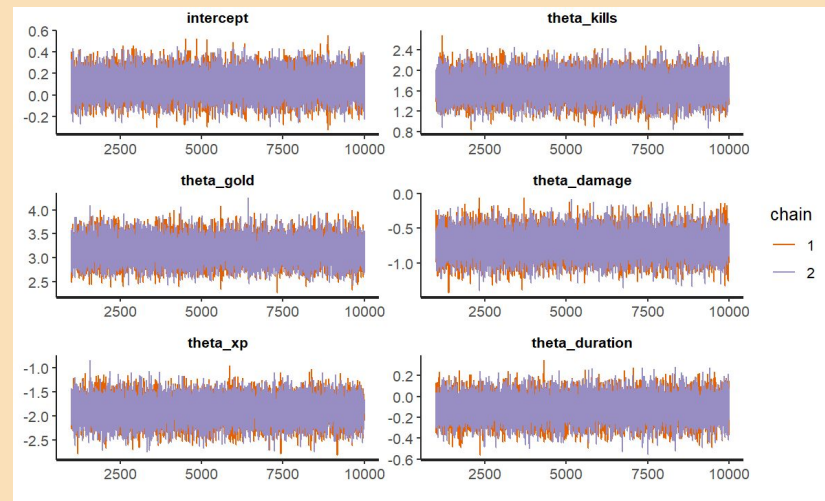
Logistic Regression Model - Group

$$\log \left(\frac{P(\text{win}_i = 1)}{1 - P(\text{win}_i = 1)} \right) = \alpha + \theta_1 \cdot \text{diff_kills}_i + \theta_2 \cdot \text{diff_gold}_i + \theta_3 \cdot \text{diff_teamfight_dmg}_i + \theta_4 \cdot \text{diff_teamfight_xp}_i + \theta_5 \cdot \text{duration}_i$$

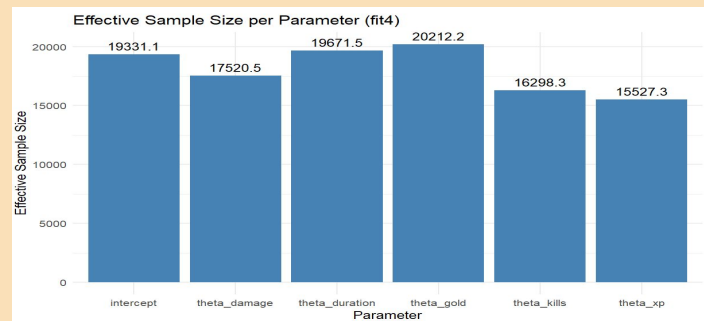
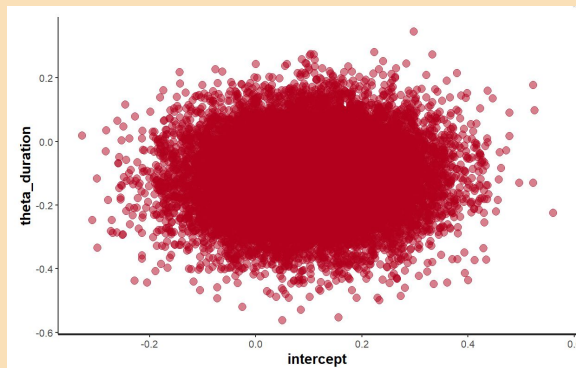
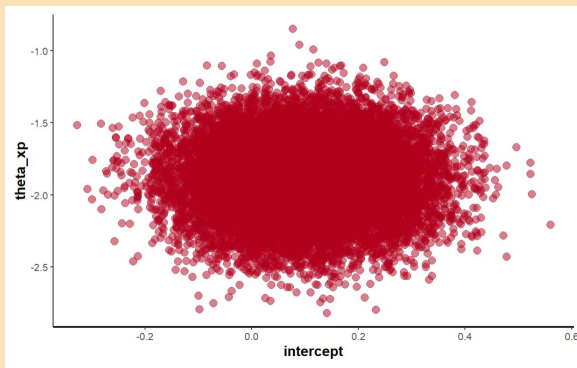
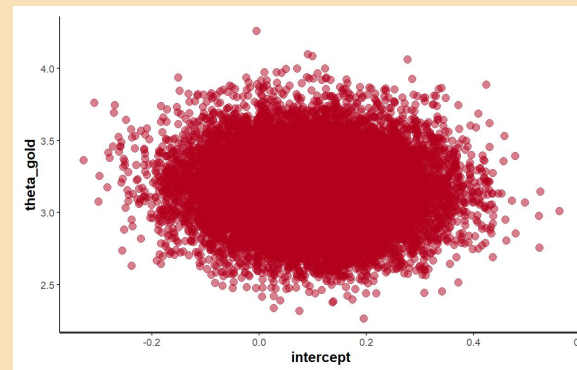
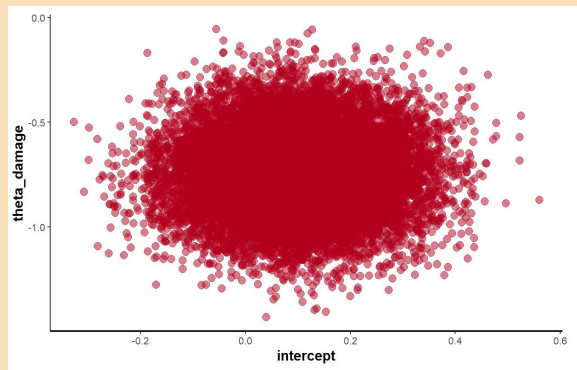
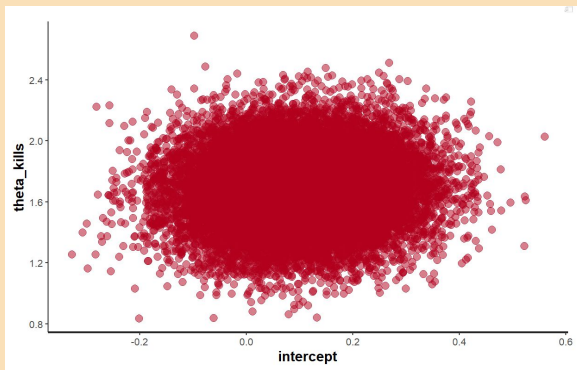
Variable and Coefficient Interpretations:

- α : Intercept
- θ_1 : Effect of **kill difference**
- θ_2 : Effect of **gold difference**
- θ_3 : Effect of **teamfight damage difference**
- θ_4 : Effect of **teamfight experience difference**
- θ_5 : Effect of **match duration**

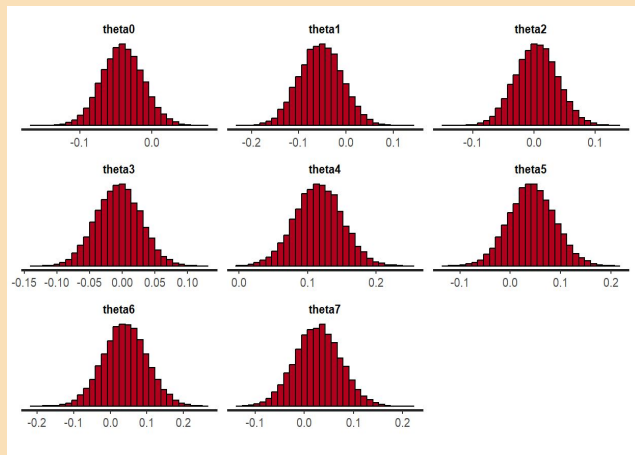
STAN: iteration = 10000, warmup = 1000, chains = 2



All coefficients have weakly informative priors, reflecting our belief that most predictors are centered around zero with moderate variability: $\theta \sim N(0, 0.5)$; $\alpha \sim N(0, 1)$



Logistics Regression - Individual



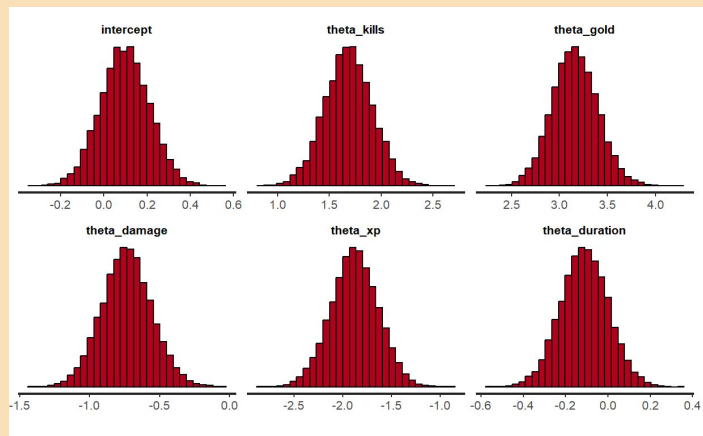
The correlation between the outcome of the game and individual performance is very small. Only gold 's 95% credible interval do not include 0.

Inference for Stan model: anon_model.
2 chains, each with iter=10000; warmup=1000; thin=1;
post-warmup draws per chain=9000, total post-warmup draws=18000.

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
theta0	-0.04	0.00	0.03	-0.10	-0.06	-0.04	-0.02	0.02	21466	1
theta1	-0.05	0.00	0.05	-0.14	-0.08	-0.05	-0.02	0.04	16144	1
theta2	0.00	0.00	0.03	-0.06	-0.02	0.00	0.03	0.07	19757	1
theta3	-0.01	0.00	0.03	-0.07	-0.03	-0.01	0.02	0.06	19828	1
theta4	0.12	0.00	0.03	0.05	0.09	0.12	0.14	0.18	20183	1
theta5	0.04	0.00	0.05	-0.05	0.01	0.04	0.07	0.13	18650	1
theta6	0.04	0.00	0.06	-0.08	0.00	0.04	0.08	0.16	16476	1
theta7	0.03	0.00	0.05	-0.06	0.00	0.03	0.06	0.12	17474	1
lp__	-3299.93	0.02	2.00	-3304.75	-3301.01	-3299.60	-3298.48	-3297.03	8168	1

Samples were drawn using NUTS(diag_e) at Sun Apr 20 11:26:09 2025.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).

Logistic Regression - Group



For group performance, the parameters show much stronger correlations with game outcomes. Based on their 95% credible intervals, we can conclude that, except for game duration, all other parameters affect the outcome.

Inference for Stan model: anon_model.
2 chains, each with iter=10000; warmup=1000; thin=1;
post-warmup draws per chain=9000, total post-warmup draws=18000.

	mean	se_mean	sd	2.5%	97.5%	n_eff	Rhat
intercept	0.10	0	0.11	-0.12	0.32	20179	1
theta_kills	1.68	0	0.23	1.23	2.13	16047	1
theta_gold	3.16	0	0.24	2.70	3.63	20339	1
theta_damage	-0.73	0	0.18	-1.09	-0.38	17474	1
theta_xp	-1.88	0	0.24	-2.36	-1.42	15185	1
theta_duration	-0.11	0	0.11	-0.33	0.11	18667	1

Samples were drawn using NUTS(diag_e) at Sun Apr 20 11:33:13 2025.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).



04

CONCLUSION





The findings from our model indicate the presence of potential unfairness within Dota 2's matchmaking system. Moreover, individual performance alone appears to have a limited impact on the final outcome of a match; success ultimately depends on coordinated team effort.

THANKS!

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